LESSON 8.1

Understand Equations and Their Solutions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an equation for a real situation and check whether a given value is a solution by substituting it.

WHY THIS MATTERS Detective Ruiz received a tip: 'A number of stolen gems plus 8 more equals 20 total gems.' She needs to write an equation to represent this clue before she can solve the case. How would you write this situation as an equation?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Equation *Spanish: Ecuación*	A math sentence with an equal sign showing both sides are the same.	x + 3 = 10
Variable *Spanish: Variable*	A letter that stands for a number that is unknown or can change.	the x in x + 3 = 10
Equal sign *Spanish: Signo igual*	The = sign, showing both sides are the same.	=
Expression *Spanish: Expresión*	A math phrase with numbers and letters, but no equal sign.	2x + 5 is an expression; it becomes an equation when you write 2x + 5 = 15
Constant *Spanish: Constante*	A fixed number in an expression or equation that does not change.	In x + 7 = 12, the numbers 7 and 12 are constants; only x can change.
Unknown *Spanish: Incógnita*	The value you are trying to find, usually shown with a variable.	In n + 5 = 12, n is the unknown; you solve to find n = 7.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A basket has b apples. You add 9 more and have 15. Is b = 6 a solution of b + 9 = 15?

1. Substitute: replace b with 6 to get 6 + 9.
2. Add: 6 + 9 = 15.
3. Compare: 15 = 15, so the equation is true.

Answer: Yes, b = 6 is a solution.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

New clue: 'A number plus 5 equals 9.' What is the unknown?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I substituted _____ for the variable and got _____ .
- The value _____ (is / is not) a solution because _____ .
- An equation is true when both sides _____ .
- The variable _____ stands for _____ in this problem.

Word bank equation | variable | solution | substitute | true | false | equal sign | both sides | balance | check

WATCH OUT!

A common mistake

A common mistake in Write Equations is writing subtraction and division clues in the wrong order

LESSON 8.1

Understand Equations and Their Solutions

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an equation for a real situation and check whether a given value is a solution by substituting it.

WHY THIS MATTERS Detective Ruiz received a tip: 'A number of stolen gems plus 8 more equals 20 total gems.' She needs to write an equation to represent this clue before she can solve the case. How would you write this situation as an equation?

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each phrase into the correct operation.

Groups: Addition (+) | Subtraction (−) | Multiplication (×) | Division (÷)

_____ more than

_____ increased by

_____ sum of

_____ less than

_____ decreased by

_____ fewer than

_____ times

_____ product of

_____ twice

_____ divided by

_____ split equally

_____ quotient of

HINT Read each phrase and think: does it mean combining, taking away, making groups, or sharing equally?

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which equation represents: 'A number plus 12 equals 30'?

(A) n + 12 = 30

(B) n − 12 = 30

(C) 12n = 30

(D) n / 12 = 30

HINT Find the operation word in the sentence. What does 'plus' tell you to do with the number and 12?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which equation represents: 'Seven less than a number is 18'?

(A) n − 7 = 18

(B) 7 − n = 18

(C) n + 7 = 18

(D) n / 7 = 18

HINT 'Seven less than a number' is tricky — the number comes FIRST, and then 7 is taken away from it.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which equation represents: 'Three times a number equals 21'?

(A) 3n = 21

(B) n + 3 = 21

(C) n − 3 = 21

(D) n / 3 = 21

HINT Circle the operation word: 'times'. What operation does 'times' always signal?

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Is n = 3 a solution of n + 5 = 9?

(A) Yes, because 3 is smaller than 9

(B) No, because 3 + 5 = 8, which does not equal 9

(C) Yes, because 3 + 5 is close to 9

(D) You cannot tell without solving first

HINT Replace n with 3.

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Which equation represents: 'A number divided by 4 equals 8'?

(A) n / 4 = 8

(B) 4n = 8

(C) n − 4 = 8

(D) n + 4 = 8

HINT The phrase 'divided by 4' names the operation. Which symbol shows division?

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you decide what expression to put on each side of the balance?

SENTENCE STARTER I put ___ on the left side because ___, and ___ on the right side because ___. The equal sign means ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 8.1

Understand Equations and Their Solutions

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an equation for a real situation and check whether a given value is a solution by substituting it.

WHY THIS MATTERS Detective Ruiz received a tip: 'A number of stolen gems plus 8 more equals 20 total gems.' She needs to write an equation to represent this clue before she can solve the case. How would you write this situation as an equation?

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each situation to its equation

_____ Item 1 Ⓐ

_____ Item 2 Ⓑ

_____ Item 3 Ⓒ

_____ Item 4 Ⓓ

2 **Complete the table. Show how you found each value.**

For each situation, identify what the variable represents and write the equation.

Situation	Variable Represents	Equation
A box holds some files. After removing 5, there are 13 left.		
A detective split 48 photos equally into folders. Each folder has 6 photos.		

SHOW YOUR WORK

3 **Decide whether each pair balances. Check Yes or No, then explain.**

Build the equation 2n + 3 = 11 on the balance scale and find n.

Left side	Right side	Balanced?

EXPLAIN YOUR CHOICE

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A detective found some fingerprints on Monday and 14 more on Tuesday, for a total of 30. Which equation represents this?

(A) f + 14 = 30

(B) f − 14 = 30

(C) 14f = 30

(D) f / 14 = 30

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Crack the Equation Error

1. **Clue** A number of case files divided by 3 equals 6.
2. **Choose a variable** Let n be the number of case files.
3. **Write the equation** 3n = 6

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did you decide what expression to put on each side of the balance?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 8.1

Understand Equations and Their Solutions

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an equation for a real situation and check whether a given value is a solution by substituting it.

WHY THIS MATTERS Detective Ruiz received a tip: 'A number of stolen gems plus 8 more equals 20 total gems.' She needs to write an equation to represent this clue before she can solve the case. How would you write this situation as an equation?

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

For each equation, explain why one situation matches and the other does not.

Equation	Situation That MATCHES	Situation That Does NOT Match
5n = 35	Five identical bags hold 35 clues total.	
n − 8 = 12	After giving away 8 items, a detective has 12 left.	

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Equation Error

1. **Situation** Twice a number is 16
2. **Identify operation** Twice means addition
3. **Write equation** n + 2 = 16

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Write a real-world situation that can be modeled by the equation n / 5 = 9. Explain what n represents and verify your equation makes sense.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How did you decide what expression to put on each side of the balance?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help